

CODSOFT ARTIFICIAL INTELLIGENCE INTERNSHIP REPORT

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Internship Domain: Artificial Intelligence

TASK 1: RULE-BASED CHATBOT

Objective: To develop a rule-based chatbot that responds to user queries using predefined conditions and pattern matching techniques.

Tools Used: • Python • Google Colab

Source Code:

```
import random
```

```
print("🤖 Smart Student Assistant")  
print("Type 'exit' to quit")
```

```
greetings = [  
    "Hello! How can I help you?",  
    "Hi there! Ask me anything.",  
    "Welcome! What would you like to know?"  
]
```

```
while True:  
    user = input("\nYou: ").lower()  
  
    if user in ["hi", "hello", "hey"]:  
        print("Bot:", random.choice(greetings))  
  
    elif "name" in user:  
        print("Bot: I am Smart Student Assistant.")  
  
    elif "course" in user:  
        print("Bot: AI & Data Science is a great field with many opportunities.")  
  
    elif "python" in user:  
        print("Bot: Python is one of the most popular programming languages.")  
  
    elif "career" in user:  
        print("Bot: Practice coding, build projects, and improve communication skills.")  
  
    elif "motivate" in user:  
        print("Bot: Success comes from consistency. Keep learning every day!")  
  
    elif "bye" in user or user == "exit":  
        print("Bot: Goodbye! Have a great day.")  
        break
```

else:

```
print("Bot: Sorry, I don't have an answer for that yet.")
```

Output:

```
 Smart Student Assistant  
Type 'exit' to quit
```

You: hello

Bot: Welcome! What would you like to know?

You: what is your name

Bot: I am Smart Student Assistant.

You: python

Bot: Python is one of the most popular programming languages.

You: Exit

Bot: Goodbye! Have a great day.

Result: The chatbot successfully responds to different user queries using predefined rules.

Conclusion: This project helped in understanding the basics of Natural Language Processing and chatbot development using Python.

TASK 2: TIC-TAC-TOE AI

Objective: To create an AI-based Tic-Tac-Toe game where a human player competes against the computer.

Tools Used: • Python • Google Colab

Source code:

```
board = [" " for _ in range(9)]
```

```
def print_board():
```

```
    print("\n")
```

```
    print(board[0] + "|" + board[1] + "|" + board[2])
```

```
    print("-+-")
```

```
    print(board[3] + "|" + board[4] + "|" + board[5])
```

```
    print("-+-")
```

```
    print(board[6] + "|" + board[7] + "|" + board[8])
```

```
    print("\n")
```

```
def check_winner(player):
```

```
    win_conditions = [
```

```
        [0, 1, 2], [3, 4, 5], [6, 7, 8],
```

```
        [0, 3, 6], [1, 4, 7], [2, 5, 8],
```

```
[0, 4, 8], [2, 4, 6]  
]
```

```
for condition in win_conditions:  
    if all(board[i] == player for i in condition):  
        return True  
return False
```

```
print("Tic-Tac-Toe Game")  
print("Positions are 0 to 8")
```

```
while True:  
    print_board()  
  
    move = int(input("Enter your move (0-8): "))  
  
    if board[move] == " "  
        board[move] = "X"  
    else:  
        print("Position already taken!")  
        continue  
  
    if check_winner("X"):  
        print_board()  
        print("You Win!")  
        break  
  
    ai_move = -1  
    for i in range(9):  
        if board[i] == " "  
            ai_move = i  
            break  
  
    if ai_move != -1:  
        board[ai_move] = "O"  
  
    if check_winner("O"):  
        print_board()  
        print("AI Wins!")  
        break  
  
    if " " not in board:  
        print_board()  
        print("Match Draw!")
```

break

Output:



... Tic-Tac-Toe Game
Positions are 0 to 8

```
| | |  
-+-+--  
| | |  
-+-+--  
| | |
```

Enter your move (0-8): 3

```
0| | |  
-+-+--  
X| | |  
-+-+--  
| | |
```

Enter your move (0-8): 2

```
0|0|X
```



...

Enter your move (0-8): 4

```
0|0|X
-+-+
X|X|O
-+-+
| |
```

Enter your move (0-8): 6

```
0|0|X
-+-+
X|X|O
-+-+
X| |
```

You Win!

Result: The game successfully allows the user to play against the AI and determines the winner correctly.

Conclusion: This project demonstrates basic AI logic, game development concepts, and decision-making algorithms.

TASK 3: NUMBER GUESSING GAME

Objective: To develop a number guessing game where the user attempts to guess a randomly generated number.

Tools Used: • Python • Google Colab

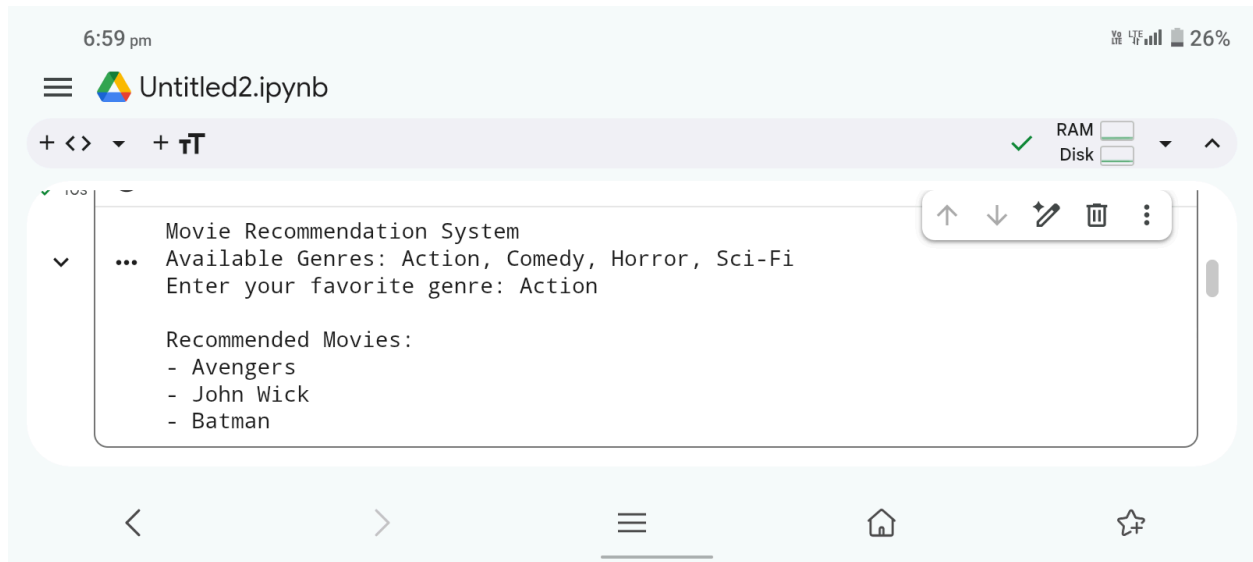
Source Code:

```
import random
```

```
def guess_the_number():  
    secret_number = random.randint(1, 20)  
    attempts = 0  
  
    print("I'm thinking of a number between 1 and 20.")  
  
    while True:  
        guess = int(input("Take a guess: "))  
        attempts += 1  
  
        if guess < secret_number:  
            print("Too low! Try again.")  
        elif guess > secret_number:  
            print("Too high! Try again.")  
        else:  
            print(f"Correct! You guessed it in {attempts} attempts.")  
            break
```

```
guess_the_number()
```

Output:



Result: The game successfully generates a random number and guides the user until the correct answer is guessed.

Conclusion: This project improved understanding of loops, conditions, functions, and random number generation in Python.

Overall Conclusion: All three projects were successfully implemented using Python in Google Colab. These projects helped in understanding chatbot development, game logic, artificial intelligence concepts, and user interaction techniques.